Feedback — Module 3

Help Center

You submitted this quiz on **Sat 25 Jul 2015 10:21 AM PDT**. You got a score of **9.00** out of **9.00**.

Question 1

The value in each edit-distance matrix element depends on its neighbors:

Your Answer		Score	Explanation
○ To the left and to the lower-left			
○ To the upper-left, to the left and to the lower-left			
○ Above, to the left, and to the right			
◉ Above, to the left, and to the upper-left	✔	1.00	
Total		1.00 / 1.00	

Question 2

Say we have filled in the approximate matching matrix and identified the minimum value (say, 2) in the bottom row. Now we would like to know the shape of the corresponding 2-edit alignment, i.e. we would like to know where the insertions, deletions and substitutions are. We use a procedure called:

Your Answer		Score	Explanation
○ Pathing			
○ Binary search			
○ Filling			
◉ Traceback	✔	1.00	
Total		1.00 / 1.00	

Question 3

Say we are using dynamic programming to find approximate occurrences of P in T. About how many dynamic programming matrix elements do we have to fill in?

Your Answer		Score	Explanation
○ |T|^2 (squared)			
○ |P|^2 (squared)			
○ |P| + |T|			
◉ |P| |T|	✔	1.00	
Total		1.00 / 1.00	

Question 4

Local alignment is different from global alignment because:

Your Answer		Score	Explanation
○ Insertions and deletions incur no penalty			
◉ It finds similarities between substrings rather than between entire strings	✔	1.00	
○ It compares three strings instead of two			
○ There is no dynamic programming algorithm for solving it			
Total		1.00 / 1.00	

Question 5

The first law of assembly says that if a prefix of read A is similar to a suffix of read B, then...

Your Answer	Score	Explanation
○ Read B might have a sequencing error at the end		

A and B might overlap in the genome ✔	1.00	
A and B should not be joined in the final assembly		
A and B must be from different genomes		
Total	1.00 / 1.00	

Question 6

The second law of assembly says that more coverage leads to...

Your Answer	Score	Explanation
more and longer overlaps between reads ✔	1.00	
more sequencing errors		
less accurate results		
Total	1.00 / 1.00	

Question 7

In an overlap graph, the nodes of the graph correspond to

Your Answer	Score	Explanation
Genomes		
Reads ✔	1.00	
Overlaps		
Bases		
Total	1.00 / 1.00	

Question 8

The overlap graph is a useful structure because:

Your Answer		Score	Explanation
A reconstruction of the genome corresponds to a path through the graph	✔	1.00	
It makes it faster to compare reads			
It helps to ignore long overlaps			
Total		1.00 / 1.00	

Question 9

Which of the following is not a reason why an overlap might contain sequence differences (i.e. might not be an exact match):

Your Answer		Score	Explanation
Insufficient coverage	✔	1.00	
Polyploidy			
Sequencing error			
Total		1.00 / 1.00	